

DHANANJAY MISHRA

Ramanujan Faculty Fellow
Dept. of MSE, IIT Kanpur

Ramanujan Faculty specializing in the interdisciplinary frontier of Non-Volatile Memory (RRAM) and Neuromorphic Computing and energy storage systems. With a proven track record in securing high-value research funding, focusing on integration of next-generation semiconductor materials and energy-efficient hardware for AI. Committed to advancing sustainable electronics through applied research in supercapacitors and 2D materials, while fostering academic excellence through dedicated student mentorship and institutional collaboration

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PROFILE SUMMARY

- Ramanujan Faculty Fellow, Dept. of MSE, IIT Kanpur in Feb 2026.
- Research Professor – KyungHee University Seoul, South Korea || Nov. 2025 – Jan 2026
- Spearheaded cutting-edge research in electronics engineering with a strong focus on semiconductor innovation, particularly in RRAM-based memory technologies and next-generation energy storage systems.
- Served as a National Research Fellow at Incheon National University, leading high-impact projects on hybrid CMOS/RRAM integration for neuromorphic computing applications.
- Demonstrated hands-on expertise in nano-fabrication processes, including photolithography, thin film deposition, and etching techniques, crucial for semiconductor prototyping and device miniaturization.
- Played a pivotal role in the successful design, fabrication, and characterization of CMOS-compatible RRAM devices, enhancing computational efficiency in in-memory and AI-centric architectures.
- Provided strategic mentorship to graduate researchers, delivering training in cleanroom protocols, advanced instrumentation, and experimental design, fostering a high-performing, knowledge-sharing research environment.
- Excelled in interdisciplinary collaboration, bridging the gap between materials science, electronics, and applied physics to develop scalable, high-performance electronic components.
- Engaged in international collaborations and co-authored multiple high-impact publications, contributing to the global advancement of memory and energy technologies.
- Leveraging a foundational background in Mechanical Engineering to solve complex thermo-kinetic and structural challenges in Nanofabrication.

EDUCATIONAL DETAILS

- M.S. + Ph.D.
Electronics Engineering
Incheon National University, South Korea | 2018 – 2023
Thesis: A Study on CuI-based Resistive Random Access Memory (RRAM) Towards Optoelectronics Applications
Advisor: Dr. Sung Hun Jin, Professor, Department of Electronics Engineering, Incheon National University
- Bachelor of Technology (B.Tech)
Mechanical Engineering
Abdul Kalam Technical University (AKTU), Uttar Pradesh, India
| 2009 – 2013

CORE COMPETENCIES

- Resistive Random Access Memory (RRAM)
- Neuromorphic Computing
- Device Modeling & Simulation (COMSOL, MATLAB)
- System level Device Integration
- 2D Materials (TMDC, MXene, CNT)
- Nanofabrication (CVD, ALD, Lithography)
- Supercapacitor and micro-supercapacitor
- Electrochemical Characterization
- Photodetectors & Sensors

NOTABLE FELLOWSHIPS & AWARDS

- Ramanujan Faculty fellow – IIT Kanpur || Feb 2026– Present
- Principal Investigator – NRF Fellowship, South Korea
Academic Balanced Development Creative Challenge Research-based Support Project (Assignment No. RS-2023-0024981)

2023 – 2026

Awarded competitive National Research Foundation (NRF) fellowship to lead an independent research project, recognizing high-impact potential in electronics engineering and neuromorphic systems.

- **Special Research Scholarship – Incheon National University, South Korea**

2018 – 2023

Granted a prestigious Ph.D. scholarship (~\$20,000/year) for outstanding academic performance and research contributions in the field of Electronics Engineering.

- **Senior Research Fellowship – Council of Scientific & Industrial Research (CSIR), India**

2018- IIT Indore

Earned through national-level selection for advanced doctoral research excellence in engineering sciences.

- **Junior Research Fellowship – CSIR, India**

2016- IIT Indore

Secured through rigorous nationwide competitive exam, supporting early-stage research initiatives in engineering and applied sciences.

- **All India Rank 75 – CSIR NET (Engineering Science)**

2015

Achieved top percentile in the CSIR National Eligibility Test, qualifying for premier research fellowships in India.

- **Cover Feature – *Advanced Functional Materials***

August 2023

Research featured on the **cover page** of *Advanced Functional Materials* journal (DOI: [10.1002/adfm.202370041](https://doi.org/10.1002/adfm.202370041)), underscoring innovation in neuromorphic devices and RRAM-based architectures.



WORK EXPERIENCE

Ramanujan Faculty | Dept. of Materials Science and Engineering, Indian Institute of Technology Kanpur, India

Feb 2026 to present:

Role:

- **Ecosystem Development:** Establishing a comprehensive "Semiconductor-to-Materials-to-Device" research environment.
- **Advanced Fabrication:** Spearheading research on freestanding CuI thin films via Pulsed Laser Deposition (PLD) for RRAM and neuromorphic applications, including BNNs and PUFs.
- **Academic Leadership:** Delivering a specialized SEMulator3D (Lam Research) course for UG and PG students, supported by the India Semiconductor Mission (ISM) and MeitY.
- **Institutional Collaboration:** Facilitating an institute-level MOU between LAM, IISc, and IIT Kanpur.
- **Mentorship:** Providing high-level guidance to Master's and PhD students on cleanroom protocols and advanced nanofabrication toolsets.

Research Professor (NRF National Research Fellow) Kyung Hee University, Seoul, South Korea

Oct 2025 – Jan 2026

Role:

- **Strategic Research Leadership:** Directed the conceptual design of cross-bar RRAM architectures, focusing on the high-level integration of BNNs and PUF security protocols for next-generation hardware
- **Academic Mentorship & Lab Management:** Oversaw the training of graduate researchers in advanced nanofabrication, establishing standard operating procedures for cleanroom efficiency and safety.

National Research Fellow NRF (Post-Doctoral) at Incheon National University, South Korea

Mar 2023 to Oct 2025:

Role:

- **Neuromorphic Computing Research:** Conducted advanced research on cross-bar Resistive Random Access Memory (RRAM) architectures for neuromorphic applications, including binary neural networks (BNNs), physical unclonable functions (PUFs), and related domains.
- **Hybrid System Integration:** Investigated and optimized the physical integration of CMOS/RRAM systems, achieving measurable improvements in energy efficiency and scalability for in-memory computing.
- **Process Engineering:** Designed and executed precise process flows for BEOL integration, ensuring all RRAM prototypes met rigorous industrial semiconductor manufacturing standards.
- **Student Mentorship & Facility Guidance:** Mentored Master's and PhD students on cleanroom protocols and fabrication procedures, ensuring safe and efficient use of advanced nanofabrication tools and research facilities.
- **Facility Mentorship:** Trained junior researchers in thin-film deposition and etching techniques within a cleanroom environment.

Senior Research Fellow at Indian Institute of Technology, Indore

Jan 2018 to Aug 2018:

Role:

- **Hydrogen Storage Research:** Conducted experimental studies on hydrogen storage using complex metal hydrides, with a focus on adsorption-desorption behavior and associated thermal characteristics.
- **Reactor Design & Pressure Studies:** Designed custom reactor beds for material testing and evaluated hydride performance under variable pressure and temperature conditions to simulate real-world storage environments.

- **Material Characterization:** Characterized the thermodynamic and kinetic properties of hydrogen storage materials, contributing to the development of more efficient and scalable energy storage systems.

Junior Research Fellow at Indian Institute of Technology, Indore
Feb 2016 to Jan 2018:

Role:

- **End-to-End Setup Development:** Designed and engineered advanced hydrogen storage experimental setups for the precise characterization of complex metal hydrides, ensuring compatibility with high-pressure and thermal analysis protocols.
- **Reactor Bed Innovation:** Developed and optimized custom reactor bed architectures to significantly enhance thermal stability, improve material cycling efficiency, and maximize hydrogen adsorption/desorption performance.
- **Hydride Behavior Optimization:** Investigated and implemented strategies to refine the performance of complex metal hydrides under varied thermodynamic conditions, contributing to more efficient and reversible hydrogen storage solutions.
- **Advanced Characterization & Analysis:** Conducted comprehensive material characterization—including kinetic studies, pressure–composition–temperature (PCT) analysis, and thermogravimetric assessments—to support the development of next-generation energy storage systems.



TECHNICAL SKILLS

Nano Fabrication Techniques:

- **Photolithography, Thin Film Deposition** techniques including:
Chemical Vapor Deposition (CVD) | Atomic Layer Deposition (ALD) | Thermal Evaporation | Magnetron Sputtering | Electron Beam Evaporation | Pulsed Laser Deposition (PLD)
- **Etching Processes:** Wet Etching, Dry Etching using Reactive Ion Etching (RIE)

Electrochemical Characterization:

- **Techniques:** Cyclic Voltammetry (CV), Galvanostatic Charge-Discharge (GCD), Electrochemical Impedance Spectroscopy (EIS), Equivalent Circuit Modeling
- **Additional Testing:** Self-Discharge, Current Leakage
- **Material & Device Work:** Synthesis of oxide, 2D materials, MXenes; 2-terminal and micro-supercapacitor device assembly

Physical and Structural Characterization:

- **Microscopy & Imaging:** SEM, TEM, AFM
- **Elemental & Structural Analysis:** EDX, XRD, XPS, FTIR, BET
- **Electrical Characterization:** I–V and C–V measurements

Software & Tools:

- **Simulation & Design:** COMSOL Multiphysics, AutoCAD, Rhino
- **Data Analysis & Visualization:** Origin, Microsoft Excel, PowerPoint, ImageJ, XPSPeak, X’Pert HighScore
- **Documentation:** LaTeX
- **Programming Languages:** Python, MATLAB



PATENTS

- **Title:** Optical-Mediated Multi-Level Flexible Copper Iodide Resistive Random Access Memory Device for Ultra-Low Power Data Storage Applications Without Forming Process and Manufacturing Method Thereof || **Status:** Filed (2023)
Inventors: Dhananjay Mishra, S.H. Jin
Patent Jurisdiction: South Korea
Patent Application No.: 10-2023-0110799
Summary: This invention proposes a CuI-based flexible RRAM device enabling low-power, multi-level non-volatile data storage, eliminating the need for a forming process, and enhancing device scalability and energy efficiency.



RESEARCH FUNDING

- **Project Title:** *Development of Transient, Energy-Efficient Halide/Oxide Heterostructure for Crossbar RRAM Array Integrated with Micro supercapacitor Toward PUF Security Applications*
Role: Principal Investigator (Ramanujan Fellowship). **Co-PI - NONE**
Duration: Feb 2026 – Feb 2031
Funding Agency: Anusandhan National Research Foundation (ANRF), India
Funding Amount: ₹ 1,19,00,000 (1.19 crore INR)
- **Project Title:** *Development of Hybrid RRAM–Micro Supercapacitor Device for Energy-Saving Edge Computing*
Role: Principal Investigator **Co-PI - NONE**
Duration: June 2023 – May 2026
Funding Agency: National Research Foundation (NRF), South Korea
Funding Amount: 210 million KRW (~₹1.35 crore INR)

Objective: To develop a multifunctional hybrid device integrating non-volatile memory with energy storage for ultra-low power edge-AI and neuromorphic applications.



PROJECTS UNDERTAKEN

Projects as Principal Investigator:

- **Development of Transient, Energy-Efficient Halide/Oxide Heterostructure for Crossbar RRAM Array Integrated with Micro supercapacitor Toward PUF Security Applications** Funding Agency: Anusandhan National Research Foundation (ANRF), India
Role: Principal Investigator (Ramanujan Fellowship)
- **Development of Hybrid RRAM-Micro Supercapacitor Device for Energy-Saving Edge Computing**
Role: Principal Investigator – Dr. Dhananjay Mishra | Duration: June 2023 – May 2026
Focused on integrating non-volatile memory and energy storage to build efficient hardware for edge-AI applications.

Completed Projects:

- **Low-Cost, High-Performance, Eco-Friendly 2D Transition Metal-Sulfide/CNT Heterojunction Supercapacitor & IoT Flexible Module Platform**
Advisor: Prof. Sung Hun Jin | In Collaboration With: Samsung
Developed scalable supercapacitor structures using green materials for integration into flexible IoT systems.
- **Korea-India Joint Research Center for Real-Time Environmental Management System**
Principal Investigator: Prof. Sung Hun Jin
A bilateral initiative to develop IoT-based environmental monitoring technologies for real-time data capture and analysis.
- **Reversible Inversion-Doping Technology for Self-Aligned Charge Active Layer-Based 2D TMDC Transistor and Photodetector**
Principal Investigator: Prof. Sung Hun Jin
Aims to develop advanced 2D material-based transistors for high-performance optoelectronic devices.
- **Flexible Dual-Perovskite Broadband Photodetector Array with Self-Powered Operation and Stability**
Principal Investigator: Prof. Sung Hun Jin
Developed flexible photodetectors capable of self-powered, wide-spectrum sensing with high operational durability.



RESEARCH INTERESTS

- **Resistive Random Access Memory (RRAM):** Design and development of CuI-based memory devices for low-power data storage.
- **Neuromorphic Hardware Applications:** Hardware implementation for PUFF and Binary Neural Networks (BNN) in next-gen computing.
- **Energy Storage Devices:** Development and integration of supercapacitors and micro-supercapacitors for compact, high-efficiency energy systems.
- **Sensors & Photodetectors:** Advanced materials and architectures for high-sensitivity and broadband optoelectronic sensing platforms.



INTERNATIONAL CONFERENCE PRESENTATIONS

- **IEEE EDTM 2024 – Electron Devices Technology & Manufacturing Conference**
March 3–6, 2024 | Bangalore, India
D. Mishra, S.G. Seo, S.H. Jin
“Tailoring SWNT Thin Film Transistor Performance: The Role of CuI Heterostructures”
Presented at the 8th IEEE EDTM Conference.
[DOI: 10.1109/EDTM58488.2024.10512327]
- **22nd International Meeting on Information Display (IMID 2022)**
August 23–26, 2022 | BEXCO, Busan, South Korea
Poster Presentation on advanced display device research using CuI-based optoelectronic materials.
- **5th International Conference on Advanced Electromaterials (ICAE 2019)**
November 5–8, 2019 | Jeju, South Korea
Oral Presentation on hybrid memory-material integration using 2D and oxide-based nanostructures.
- **6th International Conference on Advances in Energy Research (ICAER 2017)**
December 12–14, 2017 | Indian Institute of Technology Bombay, India
E. Anil Kumar, D. Mishra, V.K. Sharma
“Estimation of Enthalpy of Formation of Metal Hydrides: Effect of Different Measurement Parameters”
Published in *Advances in Energy Research*, Vol. 1 – Selected Papers from ICAER 2017.



OTHER RELEVANT INFORMATION

- **Volunteer in organizing GIAN courses at IIT Indore.**

- **GIAN certification courses on**
 1. Introduction to Thermal System Design 11th -15th Sep 2017 at IIT Indore
 2. Adsorption Science and Technology for Cooling & Desalination Application Design 11th -21st Sep 2017 at IIT Indore.
 3. Second Law Analysis of Thermal Energy Storage Systems 17th -22nd July 2017 at IIT Indore.
 4. Introduction to Heat Pipe Science and Technology 6th -19th Dec 2016 at IIT Indore.

PUBLICATIONS

- **Total Publications:** 35+ peer-reviewed journal papers
- **Top Journals:** *Advanced Functional Materials* (IF: 19), *Advanced Materials* (IF: 27.4), *Carbon* (IF: 11.6), *Journal of Materials Science & Technology* (IF: 14.3), *ACS Applied Materials Interface*, *Applied Surface Science* (8.2)

Research Domains:

- Resistive Random Access Memory (RRAM) & Neuromorphic Hardware
- Optoelectronics & Photodetectors
- Supercapacitors & Energy Storage Devices
- Hybrid Nanostructures & Heterostructures
- **Under Review:** Manuscript under preparation for submission to *Nature Electronics*

Major Publications

- **D. Mishra**, et al., "UV-Assisted SWCNT-Based On-Chip Microsupercapacitors", *Carbon*, 2024. (IF: 10.5)
- **D. Mishra**, et al., "Light-Mediated Flexible CuI RRAM for Ultra-Low Power Storage", *Adv. Funct. Mater.*, 2023. (IF: 19)
- **D. Mishra**, et al., "Tin Sulfide Hybrids Enhanced by Carbon Nanotubes", *J. Alloys Compd.*, 2024. (IF: 6.3)
- **Mishra et al.**, "Multibit, Lead-Free Cs₂SnI₆ RRAM", *Adv. Funct. Mater.*, 2024. (IF: 19)
- **D. Mishra**, et al., "Self-Discharge Mitigated CuO-NiS Supercapacitor", *J. Mat. Sci. Tech.*, 2023. (IF: 11.2)

PERSONAL DETAILS

D.O.B: 30th August 1991 || **Gender:** Male || **Nationality:** Indian || **Languages Known:** English, Hindi, Korean